The Price of Being First: Birth Order, Child Labor, and Schooling in Pakistan*

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1 Introduction

School attendance provides an incomplete picture of how families allocate children's time. Siblings who both attend school may contribute very different amounts of work to the household. Using data from Punjab, Pakistan, we study how families divide this work among siblings, and whether birth order differences in work are accompanied by differences in school attendance and adolescent depressive symptoms. We also examine how these patterns vary with household resources.

We compare children of the same mother, adjusting for age and sex. Among 159,318 children aged 5 to 17, participation in economic work declines with birth order. The gap in economic work relative to first-borns is 6.1 percentage points for third-borns and 17.1 points for children ranked fifth or later.[1] The larger gap equals about 70 percent of the first-born participation rate of 24.0 percent. Children ranked fifth or later also spend 7.3 fewer hours a week on economic work and chores, about two-thirds of the 10.8 hours that first-borns average. Attendance differs less: third-borns attend at nearly the same rate as first-borns, and those ranked fifth or later are 5.4 points more likely to attend, against a first-born rate of 79.8 percent. About half of working first-borns also attend school. The work gap between first-borns and children ranked fifth or later is 19.2 percentage points in poorer households, about twice the 9.7 points in wealthier households.

The birth order differences extend beyond work and schooling to adolescent mental health. Among 81,884 adolescents aged 10 to 17, depressive symptoms measured by the PHQ-9 decline with birth order.[2] Relative to first-borns, third-borns score 0.23 points lower on the PHQ-9 and children ranked fifth or later 0.45 points lower. The 0.45 point gap amounts to about a fifth of the first-born mean of 2.09. The probability of at least mild depressive symptoms is 2.7 percentage points lower for third-borns than for first-borns. For children ranked fifth or later, the gap is 5.6 points, about a third of the first-born rate of 16.2 percent. Boys and girls show a similar decline.

Our analysis builds on Jere Behrman's distinction between parental preferences and within-family allocations. Behrman et al. (1982) show how parents who care equally about their children may invest unequally when endowments and returns differ. Birth order differences can reflect child endowments and the effective prices of investing in different children as well as parental preferences (Behrman and Taubman, 1986). Resource scarcity also affects these choices. In rural south India, Behrman (1988) finds that parents favor older children in nutrient allocation and give less weight to equity during the lean season. Within that season, their preference for older children is weaker in

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[1] When we restrict the sample to mothers with four co-resident children of any age, the work gaps relative to first-borns are 4.0 percentage points for second-borns, 8.9 for third-borns, and 13.7 for fourth-borns. These estimates use 45,596 children and the same mother fixed effects specification. The appendix reports estimates for each family size.

[2] The nine-item Patient Health Questionnaire (PHQ-9) sums symptom responses on a 0–27 scale. Scores of at least 5 and 10 indicate at least mild and at least moderate depressive symptoms, respectively (Kroenke et al., 2001).

landholding households and in those with a higher caste rank or a more educated head. We apply this framework to children's contributions of work and accompanying differences in schooling and depressive symptoms. Our model shows how siblings' earnings and chores can generate unequal work even when parents care equally about their children; comparing poorer and wealthier households shows how this division varies with resources.[3]

Several studies examine birth order, child labor, and schooling. Edmonds (2006) relates sibling differences in child labor in Nepal to comparative advantage, while Emerson and Souza (2008) and de Haan et al. (2014) provide evidence from Brazil and Ecuador. Using mother fixed effects in India, Batheja et al. (2026) find higher grade attainment and cognitive scores among earlier-born adolescents. They also study self-esteem, self-efficacy, and chores, and find that first-borns do more chores on average. In India, child labor is associated with lower psychosocial wellbeing, including in comparisons of working and nonworking siblings (Feeny et al., 2021). In Indonesia, Jayawardana et al. (2023) link childhood work to later depressive symptoms. Within rural Chinese families, Zhang et al. (2023) find no statistically significant birth order differences in non-cognitive ability. We document how work, schooling, and depressive symptoms differ by birth order among siblings in the same population.[4]

2 A Model of Work, Schooling, and Siblings

In the model, parents care equally about two children. They choose economic work hours $m_i \geq 0$, chore hours $d_i \geq 0$, and attendance $S_i \in \{0, 1\}$ for a first-born child 1 and a second-born child 2, with ages $a_1 > a_2$ and sexes $g_i \in \{f, m\}$. They maximize

$$U = \log C + \nu \log Q - \sum_{i=1}^{2} b_i \left[\phi \left(m_i + d_i \right) + \frac{\psi}{2} \left(m_i^2 + d_i^2 \right) \right] + \sum_{i=1}^{2} S_i \left(\theta - \delta_i \right), \tag{1}$$

where $C = Y + w_1 m_1 + w_2 m_2$ is consumption, $Q = A + h_1 d_1 + h_2 d_2$ is domestic output, and $b_i = 1 + \gamma S_i$. An hour of economic work contributes $w_i = w_{g_i}(a_i) > 0$ to consumption, and an hour of chores contributes $h_i = h_{g_i}(a_i) > 0$ to domestic output. The parameters Y, A, ν, ϕ, ψ, and γ are positive. Parents give equal weight to the children and value their contributions net of their effort costs (Behrman et al., 1982; Bommier and Dubois, 2004). Attendance multiplies work costs by $1 + \gamma$. Parents cannot borrow against future child earnings or purchase domestic help. Returns rise weakly with age, with $w_f(a) < w_m(a)$ and $h_f(a) > h_m(a)$. The net value of schooling is $\theta - \delta_i$, where θ can have either sign, $\delta_i = 0$ for sons, and $\delta_i = \delta(a_i) \geq 0$ for daughters. The additional cost $\delta(a_i)$ captures restrictions on daughters' mobility associated with seclusion and may rise at secondary-school ages.

[3] Related work examines who benefits from resource transfers. Behrman et al. (2011) find that conditional cash transfers in Mexico increase schooling and reduce work among younger age groups; Aurino et al. (2023) find larger learning gains for girls and disadvantaged children from school feeding in Ghana; and Coleman et al. (2026) find broadly equal food gains across household members following transfers in Bangladesh. In Morocco, a cash transfer improves girls' school progression without reducing unpaid care (Pesando, 2026).

[4] Hou and Ahmad (2026) use the same survey to study children's work, working conditions, and depressive symptoms, including differences by sibling age rank.

For given attendance, optimal hours satisfy

$$m_i = \frac{1}{\psi}\left[\frac{\lambda w_i}{b_i} - \phi\right]_+, \qquad d_i = \frac{1}{\psi}\left[\frac{\mu h_i}{b_i} - \phi\right]_+, \tag{2}$$

where $\lambda = 1/C$, $\mu = \nu/Q$, and $[x]_+ = \max\{0, x\}$. Both children's hours enter consumption and domestic output, so the shadow values and hours are determined jointly. Strict concavity gives a unique solution. A higher return to a sibling's economic work can lower λ and the other child's economic hours; a higher return to the sibling's chores works through μ to reduce the other child's chore hours. Compare otherwise identical households in which a child of a given age, sex, and attendance is first-born in one and second-born in the other, holding the sibling's sex and attendance fixed. Because returns rise weakly with age, the second-born's older sibling is at least as productive as the first-born's younger sibling, so the second-born does no more economic work and no more chores. At the same attendance status and with positive hours, daughters' higher return to chores implies a larger reduction in their chore hours than in sons' for a given fall in μ.

More parental resources lower the value of an additional rupee and reduce the economic hours of children who work, holding attendance and productivity fixed. A child does economic work only when $\lambda w_i > b_i \phi$. In the comparison above, both the first-born and the second-born may work, only the first-born may work, or neither may work. Only the middle case produces a participation gap. Our comparison of poorer and wealthier households examines how these participation differences vary with household resources.

Parents choose attendance by comparing

$$\max\{V(0,0),\ V(1,0) + B_1,\ V(0,1) + B_2,\ V(1,1) + B_1 + B_2\}, \tag{3}$$

where $V(S_1, S_2)$ is utility from consumption, domestic output, and work costs at optimal hours, and $B_i = \theta - \delta_i$. In this comparison, schooling the child costs no more in the household where the child is second-born, because the work that schooling displaces is worth no more there. Hours adjust continuously, whereas attendance changes only when another combination becomes optimal. Work can therefore differ without a change in attendance.

3 Data and Empirical Strategy

The Punjab Child Labour Survey records children's work, hours, school attendance, and adolescent depressive symptoms.[5] Economic work in the preceding week includes paid work, unpaid work on a family farm or business, and collection of water or firewood. Total weekly hours sum economic work and domestic chores, with zeros for children who do neither. Attendance is current school attendance. We measure depressive symptoms using the PHQ-9 score and indicators for at least mild and at least moderate symptoms. The work and attendance sample contains 159,318 children of 50,120 mothers (159,316 for hours), and the PHQ-9 sample contains 81,884 adolescents of 31,544 mothers. We call households in the bottom two wealth quintiles poorer and the remaining households wealthier. A

[5] Adults report children's work, hours, and attendance; adolescents complete the PHQ-9 themselves. See the 2019–20 survey report published by the Punjab Bureau of Statistics.

child's birth order is one plus the number of the mother's older co-resident children, of either sex and any age, counted before we restrict ages for each outcome. We group birth orders 5 and above.[6]

High birth orders occur only in large families, and earlier-born children are older. We address these differences by comparing children of the same mother and controlling for age by sex:

$$y_{im} = \sum_{k=2}^{5} \tau_k \, \mathbf{1}[r_{im} = k] + A'_{im}\pi + \alpha_m + \varepsilon_{im}, \tag{4}$$

where y_{im} is the outcome of child i of mother m, r_{im} is birth order capped at five, A_{im} contains indicators for each combination of age in years and sex, and α_m is a mother fixed effect. The coefficients τ_k describe differences from first-borns. Mother fixed effects control for characteristics shared by siblings, including household wealth and family size.[7] Because mothers space their children differently, a ten-year-old can be first-born in one family and fourth-born in another. These differences in spacing separate birth order from age, assuming the same age profile for each sex across families. We use survey weights and cluster standard errors by primary sampling unit. For comparison, we estimate OLS for the same children, replacing mother fixed effects with household controls.[8]

To compare boys and girls, we add interactions between birth order and an indicator for girls:

$$y_{im} = \sum_{k=2}^{5} \tau_k \, \mathbf{1}[r_{im} = k] + \sum_{k=2}^{5} \kappa_k \, \mathbf{1}[r_{im} = k] \, Girl_{im} + A'_{im}\pi + \alpha_m + \varepsilon_{im}. \tag{5}$$

The coefficients for boys are τ_k and those for girls are $\tau_k + \kappa_k$. We test equality of their profiles by jointly testing $\kappa_2 = \cdots = \kappa_5 = 0$.[9]

4 Results

Economic work declines with birth order. Among first-borns, 24.0 percent do economic work. We estimate that participation is 2.4 percentage points lower for second-borns than for first-borns of the same mother. The gap increases to 6.1 points for third-borns, 10.9 for fourth-borns, and 17.1 for children ranked fifth or later. The 17.1 point difference amounts to about 70 percent of the first-born rate. OLS with household controls gives a smaller gap of 13.4 points for children ranked fifth or later. Including chores, first-borns work an average of 10.8 hours a week. We estimate that children ranked fifth or later work 7.3 fewer hours, a difference of about two-thirds of that mean. Figure 1 shows nearly identical economic work gaps relative to first-borns for boys and girls ranked

[6] Each outcome sample requires at least two eligible children per mother, valid maternal links, complete covariates, positive survey weights, and an observed primary sampling unit. Mothers must be at least 15 years older than their oldest co-resident child. We exclude the entire family if any two of the mother's co-resident children share an age in completed years, so that birth order is clear in each family we retain.

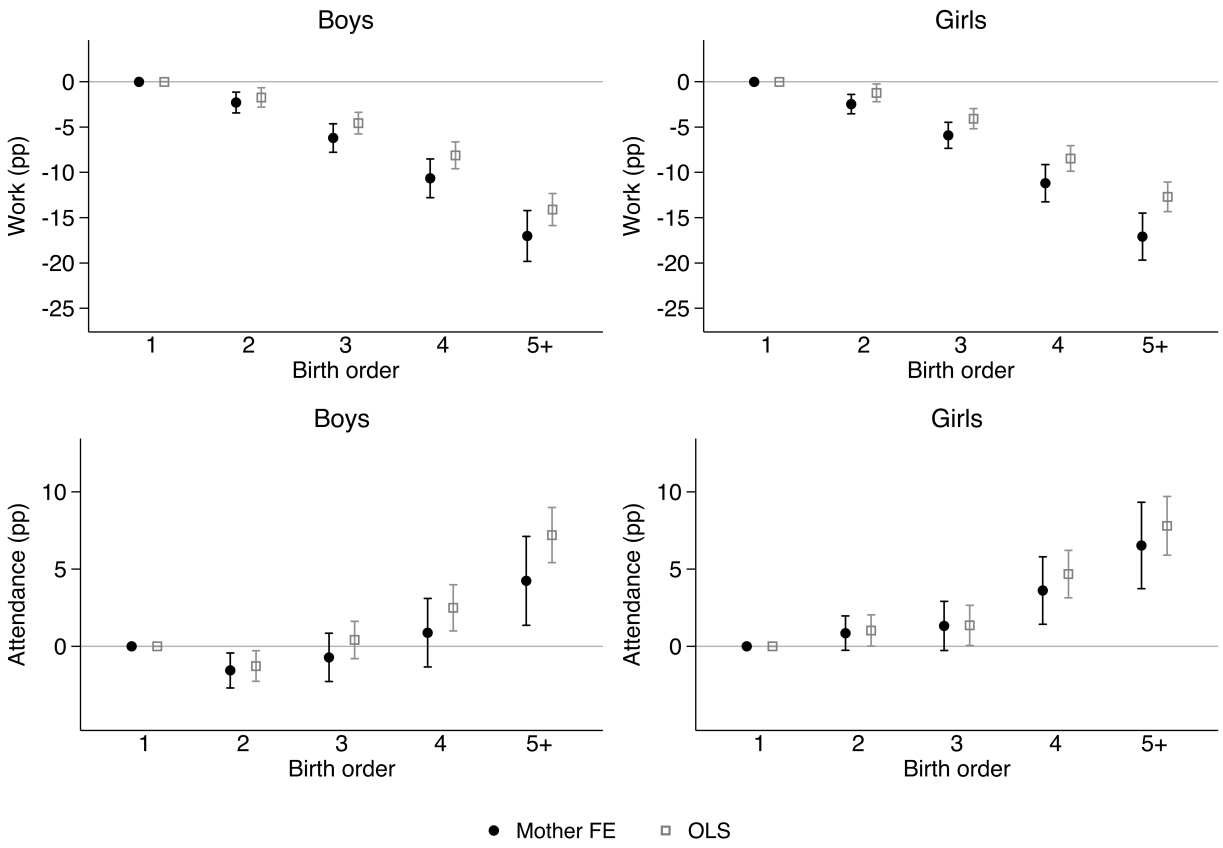
[7] To reduce the weight placed on large families, we divide each child's survey weight by the mother's number of sample children. Under these weights, children ranked fifth or later are 17.2 percentage points less likely to work and 6.2 points more likely to attend school than first-borns.

[8] The controls are the mother's schooling in five categories, her age and its square, rural residence, wealth quintile, her number of co-resident children (2, 3, 4, 5, and 6 or more), tehsil (subdistrict), and interview month.

[9] We also estimate regressions separately for boys and girls, retaining mothers with at least two eligible children of the relevant sex. In addition, we report results using relative birth order as an alternative measure in the appendix.

fifth or later (17.0 and 17.1 percentage points, respectively).[10] First-born boys are almost twice as likely to work as first-born girls (30.6 versus 16.9 percent). Chores differ more by sex. Relative to first-borns of the same sex, girls ranked fifth or later spend 3.2 fewer hours a week on chores and boys spend 1.9 fewer hours. First-born girls average 7.3 hours of chores, compared with 3.1 hours for first-born boys. We reject equal birth order profiles for boys and girls in chores but not in economic work.

Figure 1. Birth Order, Economic Work, and School Attendance



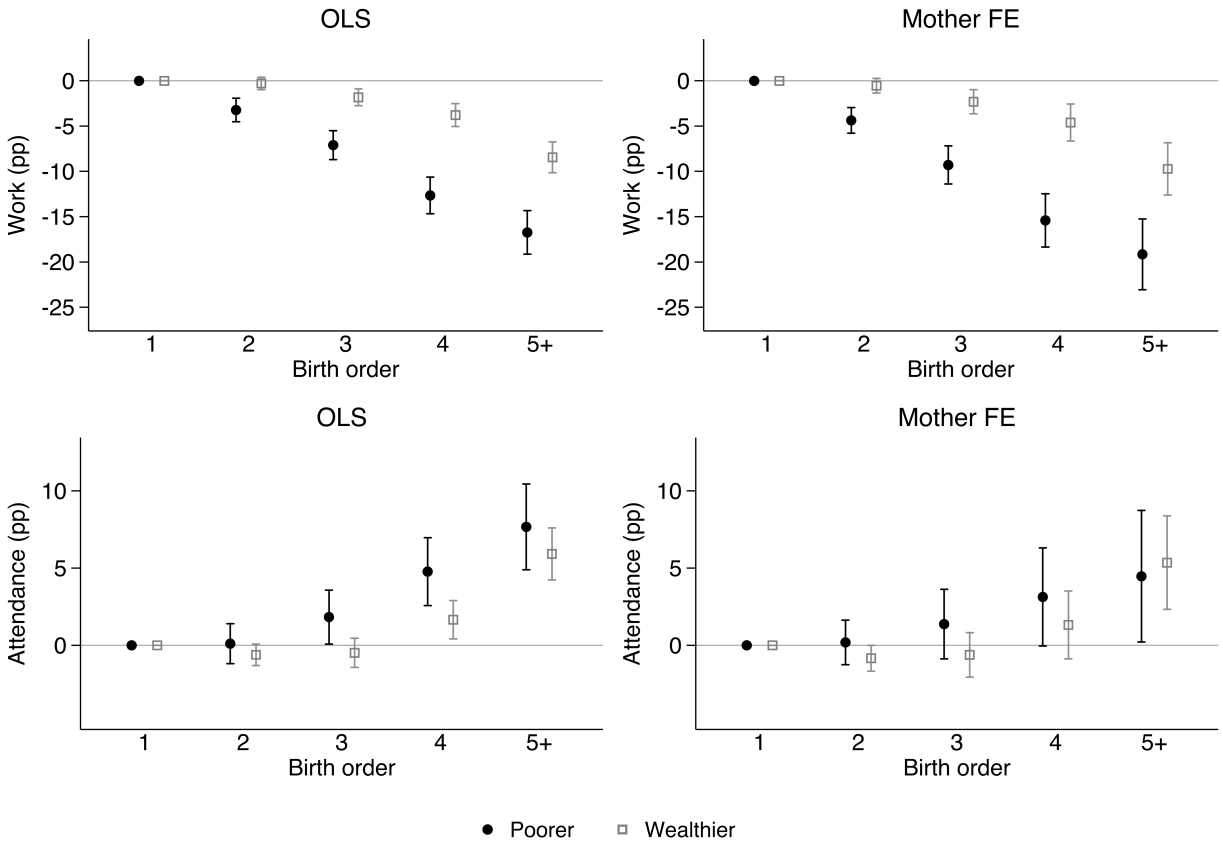
Notes: Points report $100\tau_k$ for boys and $100(\tau_k + \kappa_k)$ for girls in equation (5): differences from first-borns of the same sex in percentage points. Both specifications use 159,318 children aged 5–17 in the 2019–20 survey, survey weights, and age by sex controls. OLS replaces mother fixed effects with household controls from Section 3. Economic work includes paid work, unpaid family farm or business work, and water or firewood collection in the preceding week; attendance is current school attendance. Birth order is one plus the number of older co-resident maternal siblings of any age; we group birth orders 5 and above and normalize first-born coefficients to zero. Bars show 95 percent confidence intervals, with standard errors clustered by primary sampling unit.

School attendance differs less by birth order. Children ranked second or third attend at nearly the same rate as first-borns, and those ranked fifth or later are 5.4 percentage points more likely to attend, against a first-born rate of 79.8 percent. Many working children also attend school: 12.3 percent of first-borns combine the two, and 11.7 percent work without attending. The 17.1 point

[10]Excluding collection of water or firewood from the definition of economic work leaves the sample of 159,318 children unchanged and yields a gap of 17.2 percentage points between first-borns and children ranked fifth or later. Among the 151,469 children with both adult and child reports, estimates using either reporting source yield a gap of 17.4 points.

gap in economic work consists of an 8.5 point gap in combining work and school and an 8.6 point gap in working without attending. Attendance alone therefore understates differences in siblings' work: it treats children who combine school and work in the same way as those who attend without working. The model in Section 2 shows how a sibling's earnings or chores can reduce a child's work hours without changing school attendance.

Figure 2. Birth Order Gradients by Household Wealth

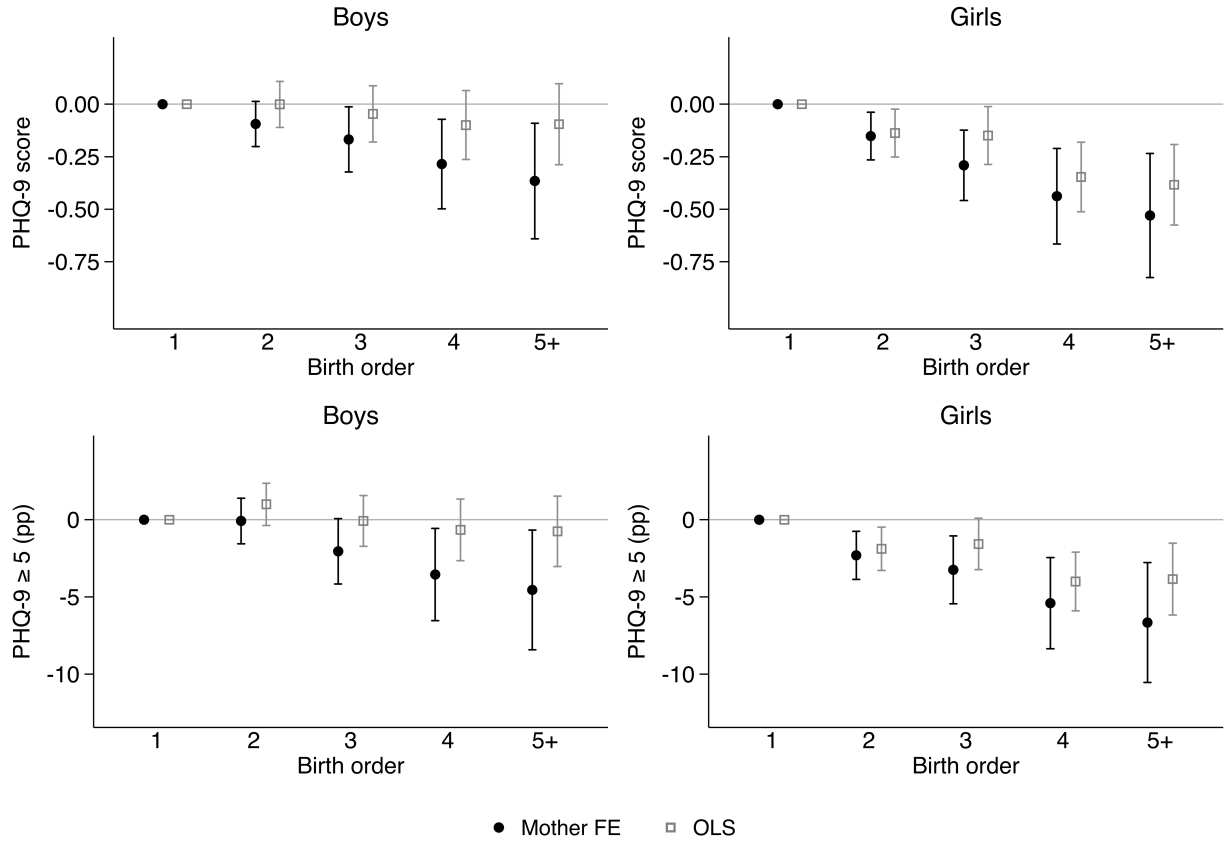


Notes: Points report differences from first-borns in percentage points (coefficients multiplied by 100), with 95 percent confidence intervals. Regressions include boys and girls aged 5–17. Poorer households are those in the bottom two wealth quintiles (57,857 children), and wealthier households are those in the top three (101,461 children). We estimate regressions separately by wealth group, using the controls, survey weights, and clustering described in Figure 1, without birth order by sex interactions. First-born coefficients are normalized to zero. Joint tests of equal birth order coefficients across groups use a model in which all coefficients differ by group. In the mother fixed effects specification, these tests give $p < 0.001$ for economic work and $p = 0.012$ for attendance.

Birth order differences in economic work are larger in poorer households. Figure 2 reports estimates separately for poorer and wealthier households. With mother fixed effects, children ranked fifth or later are 19.2 percentage points less likely to work than first-borns in poorer households and 9.7 points less likely in wealthier households. First-borns in poorer households are also far more likely to work, 37.7 versus 14.3 percent, so the gap is about half of the first-born participation rate in poorer households and about two-thirds in wealthier households. A joint test rejects equal work

profiles across the two groups.[11] The full attendance profiles also differ, although the gaps at birth order 5 or more are similar, 4.5 points in poorer households and 5.4 in wealthier households. In twelve Sub-Saharan countries, earlier-born children have less schooling than their later-born siblings in poor households but more in richer households (Tenikue and Verheyden, 2010).

Figure 3. Birth Order and Adolescent Depressive Symptoms



Notes: The top row reports differences in PHQ-9 scores; the bottom row reports differences in the probability of at least mild depressive symptoms (PHQ-9 score of 5 or more), in percentage points (coefficients multiplied by 100). Both compare children with first-borns of the same sex. The sample contains 81,884 adolescents aged 10–17 in the 2019–20 survey. The PHQ-9 sums nine self-reported symptom items on a 0–27 scale. Regressions include interactions between birth order and sex and use the specifications and survey weights in Figure 1. We group birth orders 5 and above and normalize first-born coefficients to zero. Bars show 95 percent confidence intervals based on standard errors clustered by primary sampling unit. Joint tests of equal birth order profiles for boys and girls under mother fixed effects give $p = 0.45$ for the score and $p = 0.23$ for at least mild symptoms.

Figure 3 shows that depressive symptoms are highest among first-borns and decline with birth order.[12] First-borns average 2.09 on the PHQ-9. We estimate that second-borns score 0.12 points

[11]To assess whether the wealth pattern reflects other family characteristics, we add interactions of birth order and the age by sex indicators with residence in a poorer household, maternal schooling, rural residence, average birth spacing of two years or less, a child aged 0 or 1 at home, and the father's absence. With mother fixed effects, the gap between first-borns and children ranked fifth or later is 7.4 percentage points larger in poorer households. A joint test rejects the null that all four interactions with the poorer group are zero in this sample of 159,262 children.

[12]Among the 81,788 PHQ-9 respondents with all activity measures observed, children ranked fifth or later are 15.1 percentage points less likely to work than first-borns, spend 7.6 fewer hours a week on economic work and chores, and are 9.3 points more likely to attend school.

lower than first-borns. For children ranked fifth or later, the score gap is 0.45 points, about a fifth of the first-born mean.[13] Among first-borns, 16.2 percent report at least mild depressive symptoms. Children ranked fifth or later are 5.6 percentage points less likely to report these symptoms, a difference of about a third of the first-born rate. For at least moderate depressive symptoms, the first-born rate is 3.1 percent. The estimate for children ranked fifth or later is close to zero: 0.2 percentage points higher than for first-borns (95 percent confidence interval: −1.5 to 1.9 points).

The decline is similar for boys and girls. With mother fixed effects, boys ranked fifth or later score 0.37 points lower than first-born boys and are 4.5 percentage points less likely to report at least mild depressive symptoms; for girls, the gaps are 0.53 points and 6.7 percentage points. We cannot reject equal profiles for boys and girls.

5 Conclusion

Earlier-born children contribute more work to their households, often while attending school, and earlier-born adolescents report more depressive symptoms. Measured in percentage points, the work difference is about twice as large in poorer households, where first-borns are more likely to work. In a model in which parents care equally about their children, siblings' contributions can produce unequal hours without changing attendance. These findings extend Behrman's allocation question from what parents invest in children to the work children provide and to adolescents' depressive symptoms.

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[13]The sharpened false discovery rate q-value for the score coefficient at birth order 5 or more is at most 0.001 when the 40 birth order coefficients for ten outcomes form one testing family within the mother fixed effects specifications (Anderson, 2008).

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